

Chapter 12

§12.1.

Distance Formula

$$P_1(x_1, y_1, z_1), P_2(x_2, y_2, z_2)$$

$$|P_1 P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Equation of a sphere centered at (h, k, ℓ) with radius r .

$$(x-h)^2 + (y-k)^2 + (z-\ell)^2 = r^2$$

§12.2.

Vector addition $\rightarrow$ the triangle law, the parallelogram law.

Scalar multiplication of a vector

Length or magnitude:

$$\vec{i} = \langle 1, 0, 0 \rangle$$

$$a = \langle a_1, a_2, a_3 \rangle \text{ or } a = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k} \rightarrow \langle 0, 0, 1 \rangle$$

$$\vec{j} = \langle 0, 1, 0 \rangle$$

$$|a| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Properties of vector operations.

unit vector $\rightarrow$ a vector whose magnitude is 1.

Given a vector $a \neq 0$, the unit vector that has the same direction as a is

$$u = \frac{a}{|a|}$$

§12.3.

dot-product: $a = \langle a_1, a_2, a_3 \rangle, b = \langle b_1, b_2, b_3 \rangle$

$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Properties of the dot-product

$$a \cdot a = |a|^2, \quad a \cdot b = b \cdot a, \quad a(b+c) = a \cdot b + a \cdot c, \quad (\alpha a) \cdot b = \alpha(a \cdot b) = a \cdot (\alpha b).$$

The Cosine law

Let θ be the angle between vectors a and b , then

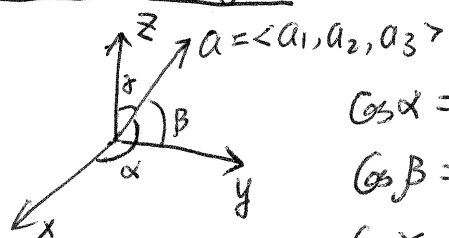
$$\cos \theta = \frac{a \cdot b}{|a||b|} \quad \text{or} \quad a \cdot b = |a||b| \cos \theta$$

i) If $a \cdot b > 0$, then $0 \leq \theta < \frac{\pi}{2}$

ii) If $a \cdot b = 0$, then $\theta = \frac{\pi}{2}$, i.e. a is orthogonal to b .

iii) If $a \cdot b < 0$, then $\frac{\pi}{2} < \theta \leq \pi$.

directional angles



$$\begin{aligned} \cos \alpha &= \frac{a_1}{|a|} \\ \cos \beta &= \frac{a_2}{|a|} \\ \cos \gamma &= \frac{a_3}{|a|} \end{aligned}$$

Projection.

$$\text{Proj}_a b = \left(\frac{a \cdot b}{|a|^2} \right) \cdot \frac{a}{|a|} = \frac{a \cdot b}{|a|^2} a \leftarrow \text{vector projection of } b \text{ onto } a$$

$$\text{Comp}_a b = \frac{a \cdot b}{|a|} \leftarrow \text{scalar projection.}$$

§12.4.

Cross-product: Let $a = \langle a_1, a_2, a_3 \rangle$, $b = \langle b_1, b_2, b_3 \rangle$

$$\begin{aligned} a \times b &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \hat{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \hat{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \hat{k} \\ &= (a_2 b_3 - a_3 b_2) \hat{i} - (a_1 b_3 - a_3 b_1) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}. \end{aligned}$$

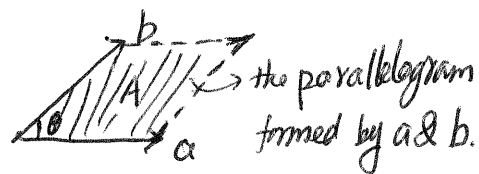
Remark: i) $a \times a = 0$.

ii) $a \times b$ is orthogonal to both a and b .

The Sine Law

p3

$$|a \times b| = |a||b| \sin \theta = \text{Area}(A) \Rightarrow$$



Remark: $a \times b = 0$ iff a and b are parallel.

Properties of the cross-product:

$$\begin{aligned} a \times b &= -b \times a, & (\alpha a) \times b &= \alpha(a \times b) = a \times (\alpha b), & a \times (b+c) &= a \times b + a \times c, \\ (a+b) \times c &= a \times c + b \times c, & a \cdot (b \times c) &= (a \times b) \cdot c, & a \times (b \times c) &= (a \cdot c)b - (a \cdot b)c. \end{aligned}$$

Triple product

$$a \cdot (b \times c) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

§12.5.

Line L : passing a point $P_0(x_0, y_0, z_0)$, direction $V = \langle a, b, c \rangle$.

Vector equation: $\langle x, y, z \rangle = \langle x_0 + ta, y_0 + tb, z_0 + tc \rangle$

parametric equations: $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$

symmetric equations: $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$.

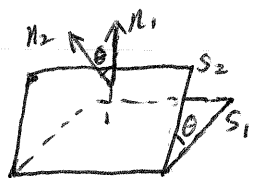
Relation of two lines:

i) Parallel ii) Intersected iii) Skew (neither parallel or intersected).

Plane S : passing a point $P_0(x_0, y_0, z_0)$, normal vector $n = \langle a, b, c \rangle$

Scalar equation: $a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$.

Angle between planes



$$\cos \theta = \frac{|n_1 \cdot n_2|}{|n_1||n_2|} \quad (0 \leq \theta \leq \frac{\pi}{2})$$

where n_1 is a normal vector of S_1 .
 n_2 is a normal vector of S_2 .

Quadratic surfaces

general form: $AX^2 + BY^2 + CZ^2 + DXY + EYZ + FXZ + GX + HY + IZ + J = 0$.

by translation and rotation, it can be converted into two standard forms:

$$AX^2 + BY^2 + CZ^2 + J = 0 \quad \text{or} \quad AX^2 + BY^2 + IZ = 0.$$

Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$

Elliptic Paraboloid: $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

Hyperbolic Paraboloid: $\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$

Cone: $\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

Hyperboloid of one sheet: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

Hyperboloid of two sheets: $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$

Chapter 13

P1

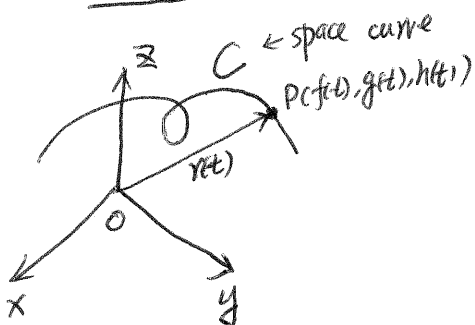
§13.1.

Vector function: $\gamma(t) = \langle f(t), g(t), h(t) \rangle$

f, g, h component functions of γ .

$$\lim_{t \rightarrow a} \gamma(t) = \langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \rangle$$

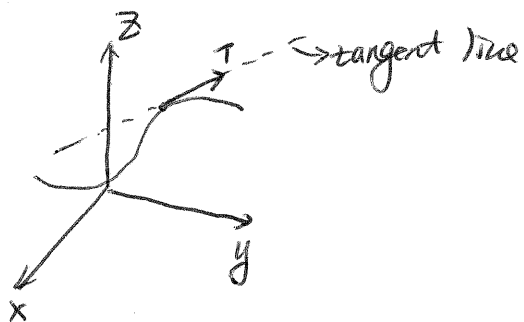
γ is continuous at a if $\lim_{t \rightarrow a} \gamma(t) = \gamma(a)$



$$x = f(t), \quad y = g(t), \quad z = h(t)$$

$\hookrightarrow$ parametric equations of the space curve C .

§13.2.



$\gamma'(t) \rightarrow$ tangent vector
 $T(t) = \frac{\gamma'(t)}{|\gamma'(t)|}$ called unit tangent vector.

Differentiation Rules

$$\frac{d}{dt} [u(t) \pm v(t)] = u'(t) \pm v'(t)$$

$$\frac{d}{dt} [\alpha u(t)] = \alpha u'(t)$$

$$\frac{d}{dt} [f(t)u(t)] = f'(t)u(t) + f(t)u'(t)$$

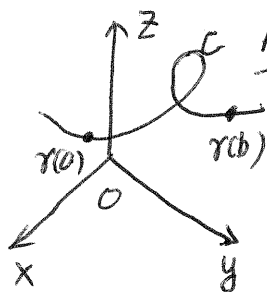
$$\frac{d}{dt} [u(t) \cdot v(t)] = u'(t) \cdot v(t) + u(t) \cdot v'(t)$$

$$\frac{d}{dt} [u(t) \times v(t)] = u'(t) \times v(t) + u(t) \times v'(t)$$

$$\frac{d}{dt} [u(f(t))] = f'(t)u'(f(t)) \rightarrow \text{Chain Rule.}$$

Theorem: If $|\gamma(t)| = C$ (a constant), then $\gamma(t) \cdot \gamma'(t) = 0$, i.e. $\gamma(t)$ is orthogonal to $\gamma'(t)$.

Integrals Let $R(t) = \int \gamma(t) dt = \langle \int f(t) dt, \int g(t) dt, \int h(t) dt \rangle$, then
 $\int_a^b \gamma(t) dt = R(b) - R(a)$.



Arc Length of the curve C from $r(a)$ to $r(b)$:

$$L = \int_a^b |r'(t)| dt$$

$$= \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2} dt$$

Arc length function: the arc length from $r(a)$ to $r(t)$

$$s(t) = \int_a^t |r'(u)| du \rightarrow \frac{ds}{dt} = |r'(t)|$$

$\rightarrow$ one-to-one function.

Curvature of a curve

$$k = \left| \frac{dT}{ds} \right| = \frac{|T'(t)|}{|r'(t)|} \rightarrow \text{formula 1}$$

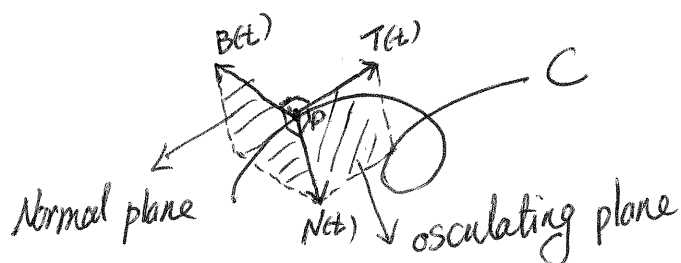
(at a general point)

or

$$k = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3} \rightarrow \text{formula 2}$$

Principal unit vector

$$N(t) = \frac{T'(t)}{|T'(t)|}$$



Binormal vector

$$B(t) = T(t) \times N(t)$$

§ 13.4.

Velocity: $v(t) = r'(t)$ where $r(t)$ is the position vector in $\mathbb{R}^3$.

$$\text{speed} = |v(t)| = |r'(t)|$$

Acceleration: $a(t) = v'(t) = r''(t)$.

$$\text{Thus } r(t) = \int v(t) dt + C_1, \quad a(t) = \int v(t) dt + C_2$$

where C_1, C_2 are determined by initial conditions.

Newton's Second Law: $F(t) = m a(t)$
 $\downarrow \quad \downarrow \quad \searrow$
 force mass acceleration.

Chapter 14

§14.1.

function of two or more variables.

$$f(x, y), f(x, y, z).$$

generally, $f(\underbrace{x_1, x_2, \dots, x_n}_{n\text{-tuple of real numbers}})$

domain: $D \in \mathbb{R}^2$

range: $\{f(x, y) \mid (x, y) \in D\}$. } for $z = f(x, y)$.

Level curves

$f(x, y) = k$ where k is a constant.

§14.2.

Limit of a function of two variables

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$$

Remark: If $f(x, y) \rightarrow L_1$ as $(x, y) \rightarrow (a, b)$ along a path C_1 and $f(x, y) \rightarrow L_2$ as $(x, y) \rightarrow (a, b)$ along another path C_2 , where $L_1 \neq L_2$, then $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ does not exist.

* f is continuous at (a, b) iff $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = f(a, b)$.

§14.3.

First partial derivatives

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \leftarrow \text{regard } y \text{ as a constant.}$$

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h} \leftarrow \text{regard } x \text{ as a constant.}$$

notations: $f_x(x, y) = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = D_x f$

$$f_y(x, y) = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y} = D_y f$$

Higher partial derivatives

$$\begin{aligned} (f_x)_x &= f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) & (f_x)_y &= f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \\ (f_y)_x &= f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) & (f_y)_y &= f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) \end{aligned} \quad \left. \vphantom{\begin{aligned} (f_x)_x &= f_{xx} \\ (f_y)_x &= f_{yx} \end{aligned}} \right\} \text{second derivatives}$$

In general

$$u = f(x_1, x_2, \dots, x_n)$$

$$\rightarrow \frac{\partial f}{\partial x_i} = f_{x_i} = D_i f = \frac{\partial u}{\partial x_i}, \quad i=1, 2, \dots, n.$$

Clairaut's Theorem

If f_{xy} and f_{yx} are both continuous at (a, b) , then $f_{xy}(a, b) = f_{yx}(a, b)$.

§ 14.5

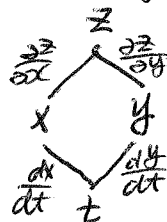
Chain Rules

Case I:

If $z = f(x, y)$ with $x = g(t)$ and $y = h(t)$, then

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = f_x g' + f_y h'}$$

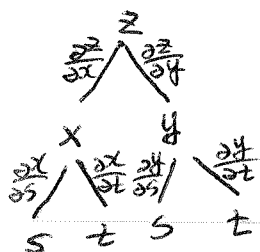
Tree diagram



Case II:

If $z = f(x, y)$ with $x = g(s, t)$ and $y = h(s, t)$, then

$$\boxed{\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} & \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} \\ &= f_x g_s + f_y h_s & &= f_x g_t + f_y h_t \end{aligned}}$$



General form

$$u(x_1, x_2, \dots, x_n), \quad x_j(t_1, t_2, \dots, t_m) \text{ for } j=1, 2, \dots, n$$

then

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i} \quad i=1, 2, \dots, m.$$